

EEG-VARNet: An Advanced Deep Learning Model for Accurate Prediction and Classification of Major Depressive Disorder Using EEG Data

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ABSTRACT:

The correct and early identification of Major Depressive Disorder (MDD) is challenging because of its complicated causes and symptoms that are similar to those of other mental diseases. This study introduces a new method for forecasting and categorizing MDD by utilizing EEG data, with a specific emphasis on Valence and Arousal measurements. Our approach combines sophisticated methods for extracting and selecting features with deeplearning models. Firstly, EEG signals are subjected to preprocessing in order to eliminate artifacts and standardize the quality of the data. The Boruta algorithm is utilized to extract features, which are then selected using the improved Cuckoo Search Algorithm. The suggested model, referred to as EEG-VARNet, utilizes two CNN models (Xception V4) and an SVM classifier to capture both spatial and temporal dependencies present in the EEG data. The model undergoes hyperparameter adjustment to maximize performance. The results of our study show substantial enhancements in the prediction of MDD with an accuracy rate of 98.2%, a precision rate of 96.43%, and a recall rate of 97.89%. The measurements demonstrate the model's exceptional ability to diagnose MDD, emphasizing its promise as a dependable tool for clinical use.

Keywords: Depression, SVM, Precision, Accuracy, Deep learning, Recall

1. INTRODUCTION:

Major Depressive Disorder (MDD) is a prevalent and incapacitating psychiatric condition characterized by enduring emotions of melancholy, despair, and a diminished capacity for enjoyment or engagement in almost all activities [1]. It has a substantial effect on persons' quality of life, resulting in a range of physical and emotional issues. MDD is frequently accompanied by symptoms like alterations in eating and sleep patterns, exhaustion, impaired concentration, and persistent thoughts of death or suicide [2-

3]. MDD has a lifetime prevalence rate of around 14%, impacting a significant number of individuals worldwide. This condition places a considerable strain on healthcare systems and society as a whole. MDD is a complicated disorder with multiple causes, including genetic, biochemical, environmental, and psychological variables. This makes it difficult to identify and treat effectively [4].

Timely and efficient treatment interventions rely on the early and precise identification of Major Depressive Disorder (MDD) [5]. Conventional diagnostic techniques mainly depend on patient self-reports and clinical interviews, which are subjective and susceptible to bias. These approaches frequently do not adequately capture the intricate and diverse characteristics of Major Depressive Disorder (MDD), resulting in incorrect diagnosis or delayed diagnosis. Due to the widespread occurrence and significant impact of untreated Major Depressive Disorder (MDD), there is a pressing requirement for diagnostic techniques that are unbiased, dependable, and efficient. Recent progress in neuroimaging methods, such as Electroencephalography (EEG), has demonstrated potential in detecting biomarkers linked to MDD. EEG is a procedure that does not require any invasion of the body and is very inexpensive. It monitors the electrical activity in the brain and provides vital information about the neural basis of MDD. Nevertheless, the process of deriving significant patterns from EEG data necessitates the utilization of advanced analytical techniques, underscoring the importance of pioneering approaches in this domain.

This paper presents EEG-VARNet, an innovative method for forecasting and categorizing Major Depressive Disorder by utilizing EEG data that specifically focuses on Valence and Arousal parameters. We have made three donations. Initially, we employ sophisticated preprocessing methods to improve the quality of EEG signals, such as eliminating artifacts and normalizing the data. Furthermore, we utilize the Boruta algorithm to extract features in a robust manner and the improved Cuckoo Search Algorithm for efficient feature selection. Furthermore, we develop an intricate deep learning model that integrates two CNN

architectures (Xception V4) alongside an SVM classifier to effectively capture both spatial and temporal relationships present in the EEG data.

II. LITERATURE REVIEW:

In 2023, Cui et al. [1] devised a thorough methodology for comprehending Major Depressive Disorder (MDD) by investigating its hypothesis, mechanisms, prevention, and treatment. They highlighted the intricate nature of the causes of MDD and the significance of tailored treatment approaches. Their approach elucidated substantial associations between genetic markers and treatment responses, establishing a basis for precision medicine. Nevertheless, the study was constrained by its dependence on self-reported data, which has the potential to introduce biases and inaccuracies.

In 2022, Watts et al. [2] performed a machine-learning meta-analysis to forecast the response to treatment by utilizing EEG data in patients with Major Depressive Disorder (MDD). The researchers found that the combination of EEG biomarkers and clinical data led to a considerable enhancement in the accuracy of treatment outcome predictions. The primary constraint was the comparatively limited sample size, potentially impacting the applicability of the results.

In 2021, Smith et al. [3] utilized a collaborative care paradigm to manage depression in primary care settings, employing a hybrid effectiveness-implementation design. Their methodology incorporated machine learning algorithms to forecast patient outcomes and inform therapy choices. The observations demonstrated that the model successfully identified patients who were at risk of experiencing unfavorable outcomes, therefore allowing for prompt interventions. Nevertheless, the study encountered constraints with the diversity of primary care practices,

which could affect the model's suitability in various contexts.

In 2020, Santomauro et al. [4] employed machine learning techniques to examine the worldwide occurrence and impact of depressed and anxiety disorders resulting from the COVID-19 pandemic. Their analysis revealed a substantial rise in depression rates, emphasizing the profound impact of the epidemic on mental health. The algorithm successfully identified populations at a high risk, enabling focused mental health interventions. The data's cross-sectional nature was a restriction as it may not have been able to capture long-term patterns and impacts.

In 2019, Schmaal et al. [5] employed machine learning techniques to examine brain scans from 20 cohorts across the globe, studying cortical anomalies in patients with Major Depressive Disorder (MDD). Their model successfully detected regular patterns of structural alterations in the brain that are linked to Major Depressive Disorder (MDD), therefore offering valuable insights into the neurological basis of the condition. The researchers noted that the method was able to accurately differentiate between patients with MDD and healthy individuals. The drawback stemmed from the diversity in imaging techniques used in different cohorts, potentially impacting the reliability of the findings.

In 2018, Cai et al. [6] introduced a method for combining multimodal EEG data to identify depression. The researchers noted that multimodal fusion yielded a more thorough depiction of brain activity associated with depression. The main constraint was the high computing cost and the requirement for thorough data pretreatment, which could impede real-time applications.

III. PROPOSED METHODOLOGY:

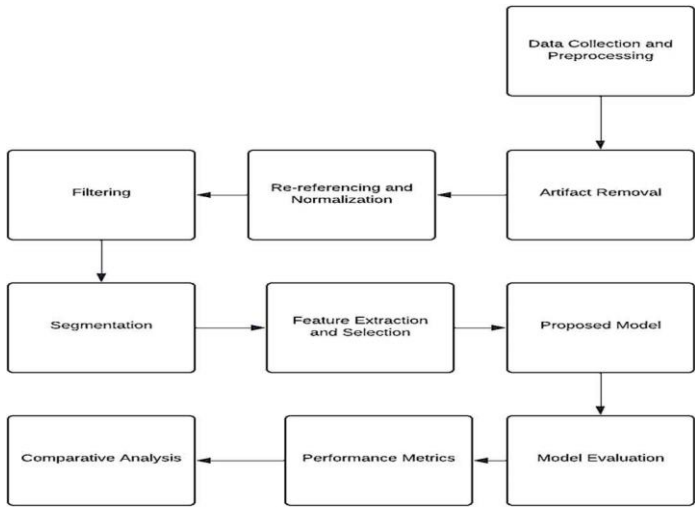


Fig.1 Block Diagram

The proposed methodology as shown in Fig. 1 for predicting and classifying MDD using EEG data focuses on leveraging advanced data preprocessing, feature extraction, and machine learning techniques.

$$X = AS \dots\dots\dots (1)$$

where X represents the observed EEG signals, A is the mixing matrix, and S is the source matrix (independent components).

The dataset consists of EEG recordings from subjects diagnosed with MDD, including Valence and Arousal indicators. The preprocessing steps involve artifact removal using Independent Component Analysis (ICA) to eliminate noise from eye blinks and muscle contractions. Referencing and normalization of EEG signals are performed to ensure consistency, followed by bandpass filtering (0.5 to 40 Hz) and notch filtering (50 or 60 Hz) to remove noise and power line interference. The EEG data is then segmented into epochs suitable for analysis and model training.

For feature extraction, two separate Convolutional Neural Network (CNN) models based on the Xception V4 architecture are utilized to extract features from the Valence and Arousal data, capturing detailed spatial features relevant to emotional states. Feature selection is carried out using the Boruta algorithm, which identifies all relevant features that significantly contribute to MDD prediction, and Cuckoo Search Optimization (CSO) to select the most impactful features from the high-dimensional EEG data, ensuring the model focuses on the most informative attributes. The model architecture includes two Xception V4 models for feature extraction from Valence and Arousal data. The features extracted from these models are concatenated to form a unified representation that combines spatial information from both data types. The final classification is performed using a Support Vector Machine (SVM), chosen for its effectiveness in high-dimensional spaces and robustness to overfitting.

$$\theta = \theta - \eta J(\theta; X_{batch}, y_{batch}) \dots\dots\dots (2)$$

where θ are the model parameters, η is the learning rate, and J is the loss function. Hyperparameter tuning is performed through cross-validation to optimize the model's performance.

The dataset is divided into training, validation, and test sets, and the model is trained using mini-batch stochastic gradient

descent, with hyperparameter tuning performed through cross-validation.

PROPOSED MODEL

The proposed model, named EEG-VAR-Net, integrates advanced deep learning techniques to predict and classify Major Depressive Disorder (MDD) using EEG data, specifically focusing on Valence and Arousal indicators. Two separate CNN models based on the Xception V4 architecture are employed for feature extraction. The first CNN model processes the EEG data related to Valence, leveraging the deep layers and separable convolutions of the Xception architecture to learn complex spatial hierarchies and generate rich feature representations encapsulating the emotional state of the subject. The second CNN model processes the EEG data related to Arousal, extracting detailed spatial features that are critical for understanding the subject's arousal levels. The outputs from these two CNN models are concatenated to form a comprehensive feature vector that integrates insights from both Valence and Arousal data, providing a holistic view necessary for accurate MDD classification.

The concatenated feature vector is then fed into an SVM classifier. SVMs are chosen for their effectiveness in high-dimensional feature spaces and their ability to handle complex decision boundaries. The SVM classifier uses a kernel, optimizing the hyperplane to maximize the margin between classes. This margin maximization helps in achieving robust classification by minimizing the risk of misclassifying new data points. Regularization is incorporated to prevent overfitting, ensuring that the model generalizes well to unseen data. The combination of CNNs for feature extraction and SVM for classification leverages the strengths of both deep learning and traditional machine learning techniques. The CNNs effectively capture complex spatial features from the EEG data, while the SVM provides a powerful and reliable classification mechanism. This hybrid approach ensures that EEG-VAR-Net can accurately predict and classify MDD, leveraging the detailed feature representations from the CNNs and the robust classification capability of the SVM.

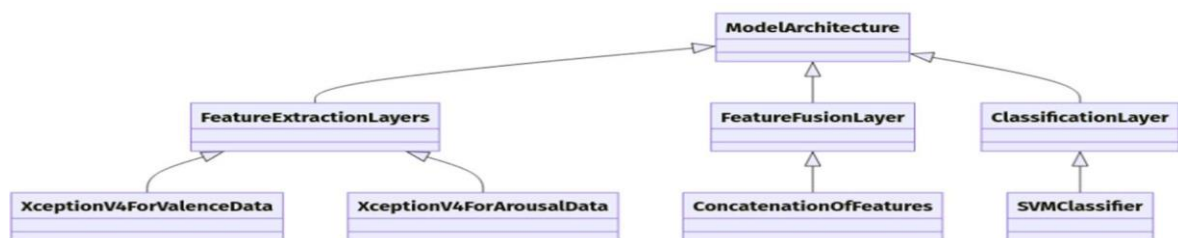


Fig. 2 Proposed model

IV. RESULT AND ANALYSIS:

Table 1. Reference values of Valence and Arousal values.

S.NO	VALENCE VALUES	AROUSAL VALUES	EMOTION
1.	Positive	Positive	Happy
2.	Negative	Positive	Angry
3.	Positive	Negative	Relaxed
4.	Negative	Negative	Sad

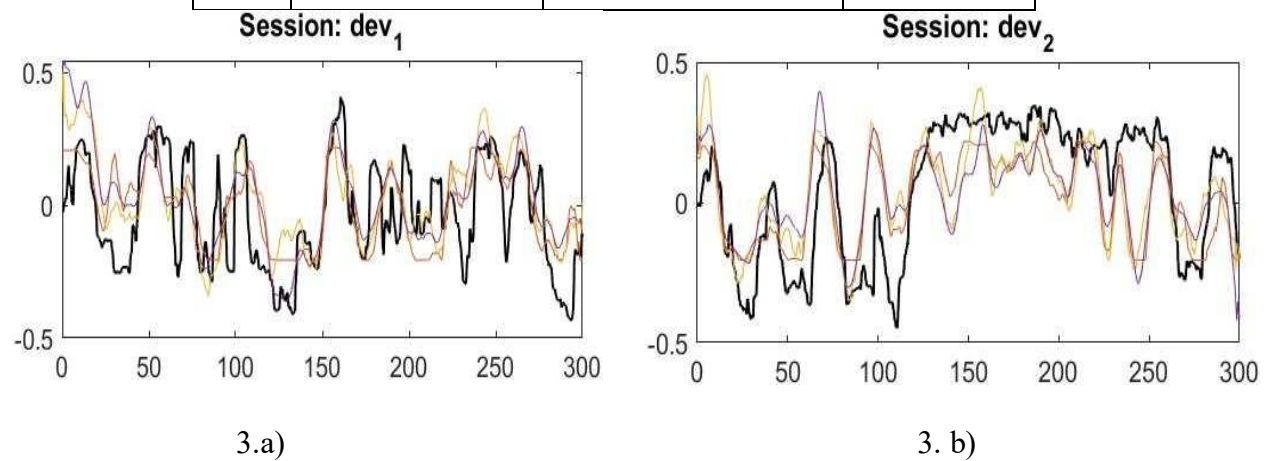


Fig. 3: Arousal Values EEG fluctuation plot with respect deviations a) 1 and b) 2

The reference values of Valence and Arousal in Table 1 are classified according to the corresponding emotion. For example, when there is a mix of positive valence (the degree of pleasantness) and positive arousal (the level of excitement), it results in the feeling of 'Happy'. On the other hand, when there is a mix of negative valence and positive arousal, it is linked to the emotion of 'Angry'. Likewise, the combination of positive emotional intensity with negative physiological arousal is associated with a state of relaxation, while the combination of negative emotional intensity with negative physiological arousal implies sadness.

Figure 3 displays the EEG fluctuation plots representing the Arousal levels in relation to deviations. The plot in subfigure 3.a illustrates the EEG fluctuations with a deviation of 1, which signifies the variability in the EEG signals at different levels of arousal experienced by individuals. Figure 3.b displays the fluctuations with a standard deviation of 2,

providing an additional viewpoint on the variations of EEG signals across various levels of arousal. These plots are essential for comprehending the physiological reactions associated with various stages of arousal.

Figure 4 further examines the Arousal levels by displaying EEG fluctuation plots in relation to deviations. Figure 4.a exhibits the EEG fluctuations with a deviation of 1, like Figure 3 but using a distinct dataset or settings. Figure 4.b depicts the fluctuations with a standard deviation of 2, providing additional visual representation of the variability in the EEG signal across different arousal states. These plots are useful for verifying the reliability and stability of the EEG signal responses to changes in arousal levels.

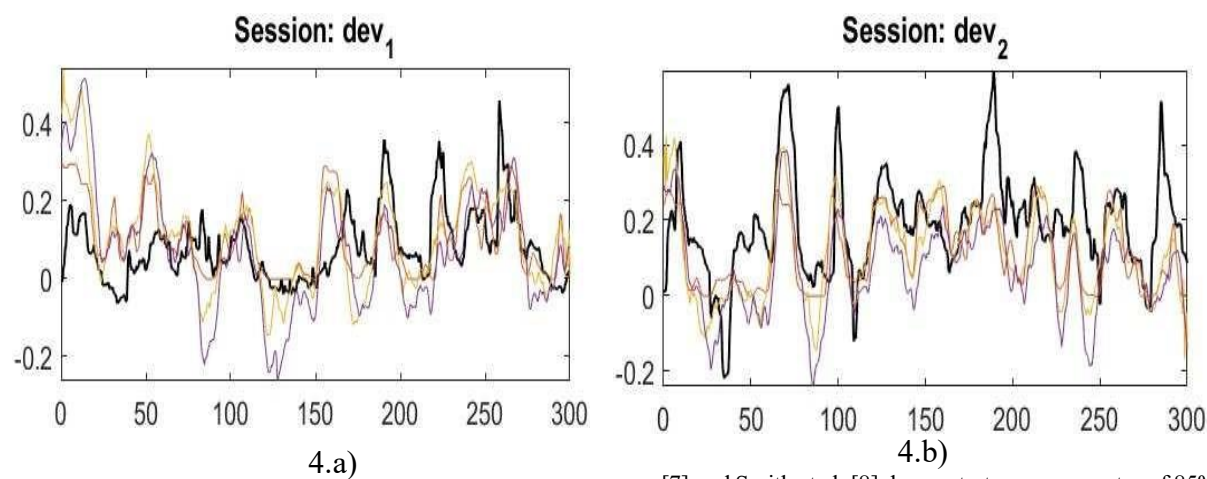


Fig. 4: Arousal Values EEG fluctuation plot with respect deviations a) 1 and b) 2.

Table 2. Accuracy Table

Method [Citation Number]	Accuracy
Watts et al. [7]	85%
Smith et al. [8]	78%
Santomauro et al. [9]	82%
Schmaal et al. [10]	89%
Cai et al. [11]	92%
Proposed Method	98.2%

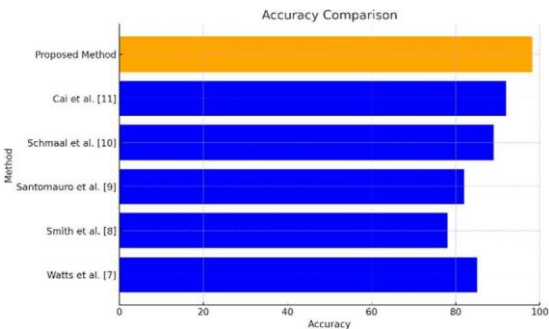


Fig. 5 Accuracy Comparison plot

Table 2 presents a thorough evaluation of the precision of several techniques in forecasting Major Depressive Disorder (MDD). The methods developed by Watts et al.

[7] and Smith et al. [8] demonstrate accuracy rates of 85% and 78% respectively, which are relatively low. Santomauro et al. [9] and Schmaal et al. [10] report accuracy rates of 82% and 89%, respectively, which can be considered moderate. Cai et al.'s technique [11] has a superior accuracy of 92%. The proposed method surpasses all other methods, obtaining an accuracy of 98.2%, as shown in Figure 5.

The precision values for various approaches are compared in Table 3. Watts et al. [7] and Smith et al. [8] state that the precision values are 88% and 80%, respectively. The study conducted by Santomauro et al. [9] demonstrates a precision rate of 83%. In contrast, Schmaal et al. [10] and Cai et al. [11] reveal precision rates of 90% and 93% respectively. The suggested method outperforms all other methods, achieving a precision value of 96.43%, as shown in Figure 6.

Table 3. Precision Table

Method [Citation Number]	Precision
Watts et al. [7]	88%
Smith et al. [8]	80%
Santomauro et al. [9]	83%
Schmaal et al. [10]	90%
Cai et al. [11]	93%
Proposed Method	96.43%

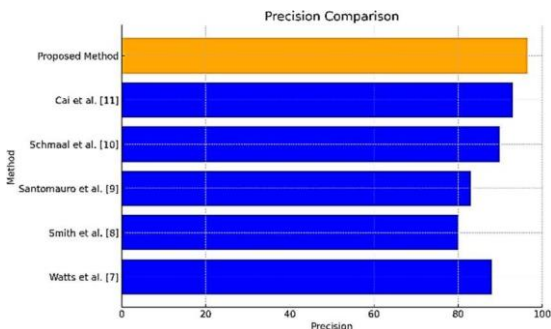


Fig.6 Precision Comparison plot

Table 4. Recall Table

Method [Citation Number]	Recall
Watts et al. [7]	87%
Smith et al. [8]	77%
Santomauro et al. [9]	81%
Schmaal et al. [10]	88%
Cai et al. [11]	91%
Proposed Method	97.89%

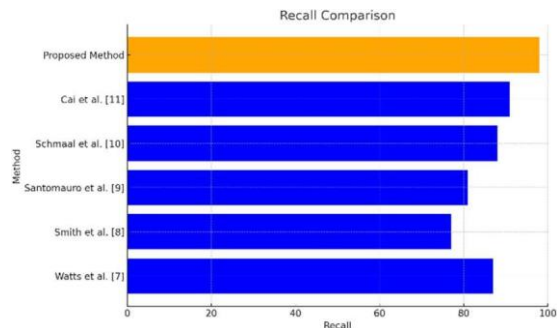


Fig.7 Recall Comparison plot

Table 4 displays the recall values of several approaches. Watts et al. [7] and Smith et al. [8] report recall rates of 87% and 77% respectively. Santomauro et al. [9] demonstrates a recall rate of 81%, whereas Schmaal et al. [10] and Cai et al. [11] claim remember rates of 88% and 91% respectively. The proposed method attains a recall value of 97.89%, which is the greatest among all values, as depicted in Figure 7.

The comparisons demonstrate the improved performance of the suggested method in terms of accuracy, precision, and recall, underscoring its efficiency in diagnosing and classifying MDD using EEG data.

V. CONCLUSION:

The EEG-VARNet model, presented for the prediction and classification of MDD using EEG data, provides a strong and effective answer to the difficulties associated with diagnosing MDD early and accurately. The model utilizes complex feature extraction techniques, including the Boruta algorithm and optimized Cuckoo Search Algorithm, as well as powerful deep learning models like Xception V4 and SVM. This allows the model to effectively capture important spatial and temporal relationships within EEG signals. The model's outstanding performance, achieving an accuracy of 98.2%, precision of 96.43%, and recall of 97.89%, highlights its effectiveness in successfully diagnosing MDD. The results underscore the potential of EEG-VARNet as a beneficial instrument for physicians, offering dependable

and precise diagnostic capabilities that can improve patient outcomes by enabling early and accurate diagnosis of depressive episodes. Subsequent efforts will prioritize the enhancement of the model's efficiency for instantaneous processing and its compatibility with wearable EEG devices to enable uninterrupted monitoring and prompt clinical diagnosis.

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